Neuron Modelling Foundation [Essential]

1) Leaky Integrate and Fire (LIF) model)

- popular tools in computational neuroscience.
- strikes a perfect balance; it's computationally "cheap" enough to simulate thousands of neurons, yet biologically realistic enough to capture how neurons process information through timing and thresholds.

- Think of it like a leaky bucket being filled with water. The water is the electric charge, and if it reaches the rim, the neuron "fires".

① Core concept

- neuron maintains an electrical potential across its membrane. In LIF model, we track this membrane potential $V(t)$ over time based on two main factors.

→ Integration: As the neuron receives inputs (synaptic current), the membrane potential increases

→ Leakage: The cell membrane isn't a perfect insulator. If no input is received, the charge slowly "leaks" out, and the potential returns to a baseline (resting) level.

→ The Fire: If $V(t)$ hits a specific threshold (V_{th}), the neuron generates a spike (action potential).

→ The Reset: Immediately after spiking, the potential is reset to lower value (V_{reset}), and the neuron enters a breif "refractory period" where it cannot fire again.

② The Mathematical Foundation

- The LIF model is described by a first-order linear differential equation. It treats the neuron membrane as an electrical circuit consisting of a resistor (R) and a capacitor (C) in parallel.

Standard Equation: $$\tau_m \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + R \cdot I(t)$$

where,

- τ_m (Membrane Time constant): Defined as $R \cdot C$. It determines how fast the "leak" happens.
- $V(t)$: The membrane potential at time t.
- V_{rest}: The resting potential (usually around $-70mV$)
- R: Membrane resistance.
- $I(t)$: Input current.

(3) Pros and cons:

Feature	Description
• Efficiency	Extremely fast to simulate, ideal for large-scale SNNs (spiking Neural Network)
• Predictability	Easy to analyze mathematically and understand the relationship betw input and output.
• Limitation	It is a 'point neuron' model - it ignores the complex geometry of dendrites and axons.
• Simplification	It doesn't model the actual ion channel dynamics (like Sodium or Potassium) seen in the more complex Hodgkin-Huxley model.

(4) Applications

- Neuromorphic computing: Designing hardware (like Intel's Loihi) that mimics brain function.
- SNNs: Training artificial neural networks that communicate via discrete pulses rather than continuous members.
- Brain Mapping: Simulating how different regions of the brain interact at a high level.

(5) Limitations

Feature	Biological Neuron	LIF Neuron Model.
• spike shape	Dynamic, carries information.	None (instant reset)
• Dendrites.	Complex spatial filtering	None (point-summation)
• Metabolic state	changes based on energy/fatique	Infinite "energy", no fatique.
• Stochasticity	Noisy and probabilistic	Deterministic (unless noise is added)

Izhikevich Model

- If the LIF model is a "Leaky bucket", then Izhikevich Model is a "flexible chameleon".
- Developed by Eugene Izhikevich in 2003, this model is the gold standard for researchers who need the computational efficiency of the LIF model but biologically "richness" of the HODGKIN-HUXLEY model. It can replicate almost any firing pattern found in brain (bursting, chattering, adapting) using just two simple differential equation.

(1) The Mathematics

- The model uses two variables: v (membrane potential)
  u (recovery variable that accounts for activation of K^+ ionic currents and inactivation of Na^+ channels).

Equation:

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

$$\frac{du}{dt} = a(bv - u)$$

Reset Logic:

When the potential reaches its peak (usually +30 mV), the variables are reset:

- If $v \geq 30$ mV, then $\{ v \leftarrow c, u \leftarrow u + d \}$

② The four Parameters

- The "~~mgt~~ magic" of Izhikevich model lies in its four dimensionless parameters (a, b, c, d). By tweaking these, you change the personality of neuron:

  - **a**: the time scale of recovery variable u. Smaller values result in slower recovery.
  - **b**: The sensitivity of u to the fluctuations of V.
  - **c**: The after-spike reset value of V (caused by fast-activated K+ channels)
  - **d**: After spike reset of u (caused by slow-activated Na+ & K+ channels)

③ Why it Matters

- The Izhikevich model is powerhouse in the field because of its versatility. While a Hodgkin-Huxley neuron might require hundreds of floating-point operations to simulate one millisecond, the izhikevich model requires only about 13.
- This efficiency allows researchers to simulate tens of thousands of neurons in real-time on a standard PC, making it the go-to choice for large scale neural network simulation.

④ Common firing Behaviors.

- The model can be turned to represent different types of cortical neurons. Here are the most famous configurations:

Neuron Type	Pattern Description	Typical Parameter.
• Regular spiking (RS)	Fires a few spikes with short intervals, then frequency decreases.	a = 0.02 b = 0.2 c = -65 d = 8
• Fast spiking (FS)	High frequency firing with slowing down (often inhibitory)	a = 0.1 b = 0.2 c = -65 d = 2

· Chattering (CH)	Fires high-frequency bursts of spikes	$a = 0.02$ $b = 0.2$ $c = -50$ $d = 2$
· Intrinsically Bursting (IB)	Fires a burst a spikes followed by single spikes	$a = 0.02$ $b = 0.2$ $c = -55$ $d = 4$

(5) Why use It?

The Izhikevich model is essentially a 2D system.

- Computational Cost: It requires only about 13 floating-point operations per millisecond of simulation.
- Biological Realism: Unlike the LIF model, it can show "resonance" (responding better to specific input frequencies) and "rebound" (firing after a period of inhibition).

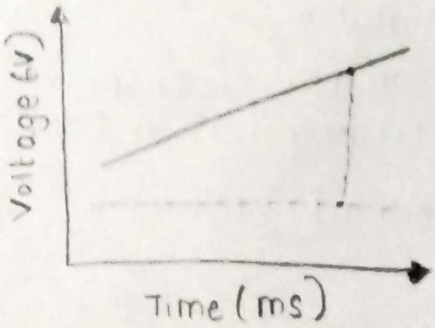
Comparison of Models.

Feature	Leaky Integrate and Fire (LIF)	Izhikevich model	Hodgkin-Huxley (HH)
· Complexity	very low (1D equation)	Medium (2D eqn)	High (4+ Differential Eqn)
· Biophysical Realism	low (Abstraction)	Moderate (Phenomenological)	very High (Ion channels)
· Computational cost	Minimal	low	very high
· Diverse Firing Patterns.	No (Mostly "Regular")	Yes (Bursting, chattering, etc).	Yes (Highly detailed)
· Best Use Case.	Large-scale SNNs / AI hardware	System Neuro-Science Brain dynamics	Detailed single cell research.

Quick visualization

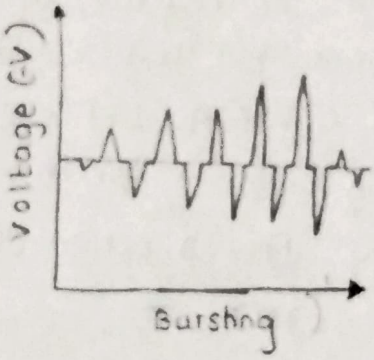
- LIF is a straight line towards a wall (threshold)
- Izhikevich is a parabolic flow that allows for "loops" and "spirals", which is why it can produce bursts and oscillations
- Hodgkin-Huxley is a high-dimensional landscape of interlocking gears

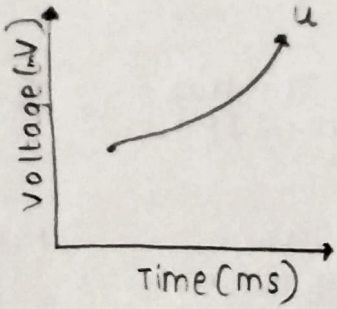
Integrate and fire



- simple equation
- Fastest simulation
- ~~Fastest~~ on/off switch behavior
- Great for large network (timing).

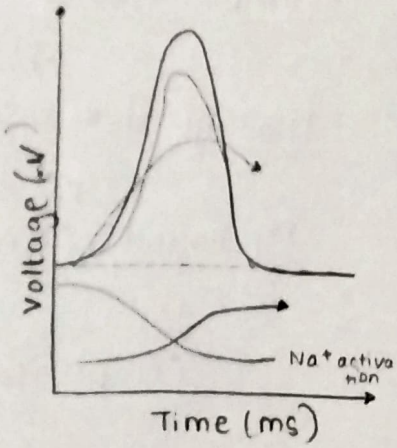
Izhikevich





- Two equation.
- Diverse firing pattern. (Regular, Bursting, chattering)
- Requires 4 parameter (a, b, c, d)

Hodgkin-Huxley



- Four + Differential Equations.
- Slowest simulation.
- Biologically accurate (Ion channels)
- Great for detailed cell studies.

Quick Questions:

1) How does a network learn a task? → Stick with LIF
2) How do different types of brain cell interact? → Izhikevich
3) How does a specific mutation or drug change neuron behavior? → Hodgkin-Huxley